MIL-S-36157E
19 November 1975
SUPERSEDING
MIL-S-36157D
2 April 1974 and
Amendment-4
25 February 1972

MILITARY SPECIFICATION

SYRINGES (AND NEEDLES), HYPODERMIC, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable, plastic syringes, with and without needles, suitable for use for a single injection of hypodermic medication.

1.2 Classification. Syringes shall be of the following types and sizes as specified (see 6.1):

Type I. - Syringe without needle.
Type II. - Syringe with needle.
Size 1. - 1 cubic centimeter.
Style a - regular
Style b - long
Size 2. - 3 cubic centimeters.
Size 3. - 5 cubic centimeters.
Size 4. - 10 cubic centimeters.
Size 5. - 20 cubic centimeters.
Size 6. - 30 cubic centimeters.
Size 7. - 50 cubic centimeters.

2. APPLICABLE DOCUMENTS

2. Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

MMM-A-250	—	Adhesive, Water-Resistant (for sealing fiberboard boxes).
PPP-B-566	—	Boxes, Folding, Paperboard.
PPP-B-585	—	Boxes, Wood, Wirebound.

FSC 6515

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PPP-B-601 _ Boxes, Wood, Cleated-Plywood.
PPP-B-621 _ Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 _ Boxes, Shipping Fiberboard.
PPP-B-676 _ Boxes, Set-Up.
PPP-T-76 _ Tape, Pressure-Sensitive Adhesive Paper (for carton sealing).

Military

MIL-N-36124_ Needles, Hypodermic, Disposable.
MIL-S-36586_ Sterilization Test Strip, Bacterial Spore.

STANDARDS

Military

MIL-STD-105_ Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129_ Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z70.1 - American Standard Dimensions of Glass and Metal Luer Tapers For Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

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U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act (FDA) and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using agencies.)

3. REQUIREMENTS

* 3.1 Material. All material shall comply with the requirements of the Federal Food, Drug, and Cosmetic Act. The portions of the unit that come in contact with the fluid to be injected shall be compatible with all known hypodermic medications. Also, the plastic components of the syringe shall comply with USP requirements for Class III plastic containers.

* 3.1.1 Barrel. Syringe barrels shall be fabricated from rigid, transparent, virgin, nontoxic, high-grade polypropylene plastic (see 3.1). Selected regrind free from graduation filler and any other adulterants may be used. The barrel shall be considered to be transparent if the scale can be seen in daylight by the naked eye when viewed through the empty barrel from the side opposite to the scale.

3.1.1.1 Alcohol resistance of graduation markings. The graduation markings shall withstand vigorous rubbing of an alcohol moistened cotton swab and remain legible when tested according to 4.4.7.

* 3.1.2 Plunger. The plunger rod shall be fabricated from rigid polypropylene or polystyrene plastic. The plunger tip shall be fabricated from fresh stock, pharmaceutical grade, nonblooming, nonleaching, nontoxic, age-resistant, inert, natural rubber, or neoprene (see 3.1).

* 3.1.2.1 Inertness to medication. The plunger tip shall show no evidence of a reaction with a high pH buffer solution when tested according to 4.4.8.

3.1.3 Needle. Needle shall be fabricated of the materials specified in MIL-N-36124.

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3.1.4 Lubricant (coating). Shall be inert, nontoxic, silicone compound suitable for the specific use. It shall also serve as a siliconizing coating on all interior surfaces of the syringe that come in contact with the plunger tip, while being used for making an injection. Silicone compound shall be a pharmaceutical grade (see 3.1).

3.2 Style, design and construction. Shall be a sterile, pyrogen-free, nontoxic, disposable, hypodermic syringe (with or without a needle, as specified). The syringe shall consist of a plunger, barrel, and needle (when specified); suitable for administering a single injection of hypodermic medication. Type I syringe shall be supplied without a needle. Type II syringe shall be supplied with a needle, of the gage and length specified (see 6.1) attached to the barrel and ready for use (see 3.2.3).

* 3.2.1 Barrel. The syringe barrel shall be fabricated from plastic, as specified in 3.1.2. The barrel shall consist of a tube closed at one end, except for an orifice, and open at the other end to receive the plunger. At the closed end, there shall be formed a tapered tip, which shall have a 0.060 inch taper per inch, to receive a standard hypodermic needle. The tip shall be formed concentric to the axis of the barrel, and shall make a mechanically stable, fluid tight connection (see 3.4.2) with a hypodermic needle having a Luer hub with hub cavity in accordance with Figure 2 (female end) of American National Standards Institute (ANSI) Standard Z70.1. The barrel may be provided with a threaded female Luer lock or have a plain slip fit tip. The female thread of the Luer lock shall be so made as not to restrict the advance of the needle hub to a leak-tight fit on the taper of the Luer tip. The Size 7 barrel, however, shall be provided with a threaded female Luer lock. The tip at the distal end shall have an outside diameter of 0.155 to 0.160 inch. The outside diameter of the tip at an altitude of 0.250 inch from the distal end shall be 0.170 to 0.175 inch. The length of the tip, except for a possible undercut or cylindrical cut at the junction with the barrel, may be greater than 0.250 inch providing the same 0.060 inch/inch taper is maintained. The wall at the open end of the barrel shall extent radially outward to form a flange of suitable size and shape to provide finger grip and anti-roll features. The inside of the barrel shall have a very fine, uniform surface, substantially free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect the fluid tightness and smoothness of its fit with the plunger. The inside surface of the barrel shall be lightly coated with the silicone lubricant specified in 3.1.4. Dimensions of the barrel shall be as specified in table I. Inside diameter shall be such as to provide accurate fit with plunger tip, resulting in a fluid-tight fit (see 3.2.2 and 3.4.1), and yet permitting plunger to move freely throughout its excursion in the barrel. Ungraduated portion of barrel, at proximal end, shall be of sufficient length to permit plunger tip to be aligned with top graduation mark, while barrel is encompassing entire rubber tip, but shall be not greater than one-third of the overall length of the syringe barrel.

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Table I. Barrel dimensions and graduations

Size	Overall length (inches)1/	Outside diameter (inches)2/	Wall thickness minimum (inches)	Orifice diameter (inches)	Sub-graduations	Main graduations	Numbering
1(1cc), style a	2-1/2 to 4	0.2 to 0.4	0.020	0.015 to 0.080	0.01cc and 0.5M#	0.1cc and 1.0M#	0.1cc and 4.0M#
1 (1cc), style b	2-3/4 to 4	0.2 to 0.3	0.020	0.015 to 0.080	1 unit and 2 units (100)	5 and 10 units	Each 10th unit
2 (3cc)	2-7/8 ± 3/8	0.4 to 0.5	0.030	0.030 to 0.080	0.1cc	0.5cc	Each 0.5cc
3 (5cc)	2-3/4 to 3-1/2	0.5 to 0.75	0.035	0.030 to 0.080	0.2cc	1.0cc	Each 1.0cc
4 (10cc)	3-1/2 to 4-3/8	0.6 to 0.85	0.036	0.030 to 0.080	0.2cc	1.0cc	Each 1 or 2cc
5 (20cc)	4-3/8 ± 1/8	29/32 ± 1/32	0.036	0.030 to 0.080	1cc	5cc	Each 5cc
6 (30cc)	4-9/16 ± 5/16	1-1/16 ± 3/32	0.040	0.030 to 0.082	1cc	5cc	Each 5cc
7 (50cc)	5-15/32 to 6-1/8	1.15 to 1.22	0.042	0.030 to 0.082	5cc (1/4 Oz)	10cc	Each 10cc (Each 1/2 oz)

1/ Including tip, without needle.
2/ Measured near distal end.
M = Minims

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* 3.2.1.1 Graduations and other markings. The outside surfaces of the barrel shall be marked with a graduation scale and the name or registered trademark of the contractor. The graduation scale shall be applied to the barrel in such a manner as to be approximately in line with the flattened side of the flange. The zero (0) mark of the scale shall register precisely with the distal ring of the plunger tip, when plunger is pushed to the bottom of the barrel. The scale and numbering shall be in cubic centimeters as specified in Table I. Main graduation marks shall be at least 33-1/3 percent longer than subgraduation marks. Except that on size 2 syringes, the last 1/2 cc and 1 cc, respectively, may be offset, or may be graduated with dots or even-sized lines. No number is required at the zero (0) mark. The markings shall be legible and of a dark, contrasting color, easily readable. On size 1, style b syringes, the barrel shall bear the word "UNITS" immediately above the numeral 100. The legend "For use with U-100 Insulin only" or similar words shall be marked adjacent to the scale.

* 3.2.2 Plunger. The plunger shall be either hollow or solid, fabricated from the material specified in 3.1.3. The distal end of the plunger shall be shaped to permit attachment of a plunger tip. The proximal end of the plunger shall extend well out of the barrel, and shall be shaped to form a knob of thickness and diameter appropriate to the size of the syringe, and shaped to allow a suitable finger grip for usage. The distal end of each plunger with its attached plunger tip shall accurately fit its individual barrel so as to make the syringe successfully withstand the fluid tightness requirements specified in 3.4.1, permit the plunger and plunger tip to draw liquids, and yet move freely and smoothly throughout its excursion in the barrel. The plunger tip shall be securely attached to the plunger rod so as not to become loose or detached when in use. The tip shall be shaped to closely fit the distal end of the barrel so as to remove all fluid from the inside of the barrel when the plunger is pushed all the way in. In addition, the tip shall be designed, e.g., beveled, in order to permit easy and efficient removal of air bubbles from fluid path. The plunger tip shall be of lubricant that detracts from appearance, interferes with visibility of plunger tip ring, or can be caused to form pools or droplets, or can be washed out with the solution being used, shall be cause for rejection. Graduation markings, denoting correct, accurate capacity, shall register precisely with the first ring at the plunger tip distal end, touching sides of the barrel. Tip shall be of a permanent, suitable color so that the first ring at the plunger tip distal end, touching the sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall. Each plunger shall have a suitable solid thumbrest at its proximal end. Thumbrest shall be perpendicular to the centerline of the plunger and may be flat or slightly curved. As an alternate, thumbrest may have a centered hole which shall not exceed 3/8 inch in diameter. Thumbrest shall be fabricated from material as specified in 3.1.2.

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* 3.2.3 Needle (type II syringe only). Each type II syringe shall be supplied with a needle fitted onto the barrel tip, ready for use. The needle shall be sufficiently set to prevent its dislodgement during shipment, handling, and use. The needle shall be of the gage and length specified (see 6.1), and shall be in accordance with the requirements of MIL-N-36124, except that as an alternate, hub shall be Luer slip in lieu of Luer lock. As an alternate, the needle may be permanently attached to the syringe. In such a case there is no Luer taper gaging required. Fluid tightness requirements shall still apply.

3.2.4 Tip protector. Each type II syringe shall be provided with a protector fabricated from a suitable rigid, non-toxic, high grade plastic. The protector shall fit securely over the syringe tip, and/or needle hub, covering the needle cannula, and shall maintain the sterility of the needle and syringe tip. The protector shall be so secured on the syringe as to remain in place during shipment and normal handling. Tip protectors without sealed ends shall contain sufficient cotton to maintain the required sterility. Each Type I syringe shall be provided with a tip protector fabricated from a suitable rigid, non-toxic, high grade plastic. The tip protector shall fit securely over the syringe tip. As an alternate, the entire syringe may be encased in a suitable, rigid, non-toxic, high grade plastic tube with cap designed to maintain sterility and provide means of dispensing the syringe by sterile technique (see 5.1.3.1).

3.3 Accuracy. When tested as specified in 4.4.1, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.4 Fluid tightness.

* 3.4.1 Fluid tightness between plunger and barrel. The syringe shall be tested for fluid tightness between the plunger and barrel, as specified in 4.4.2.1, 4.4.2.2 and 4.4.2.3.

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3.4.2 Fluid tightness, syringe tip to hypodermic needle hub. The syringe tip shall be tested for fluid tightness between the tip and hypodermic needle hub, as specified in 4.4.2.1, 4.4.2.2 and 4.4.2.3.

3.4.3 Suction. There shall be no air leakage as revealed by the suction test as specified in 4.4.2.2 or in 4.4.2.3.

3.5 Freedom from foreign matter. When tested as specified in 4.4.3, the fluid path of the syringe, and needle for type II syringes, shall be free from chips, lint, dirt, or other foreign matter.

3.6 Safety (toxicity) and lead content.

3.6.1 Safety (toxicity). When tested as specified in 4.4.4.1, the syringes, including tip protectors, shall comply with safety test requirements of the Pharmacopeia of the United States (USP).

3.6.2 Lead content. When tested as specified in 4.4.4.2 or 4.4.4.3, all plastic and rubber components coming in contact with the medication being administered shall contain not more than 5 parts per million (p.p.m.) of lead.

3.7 Freedom from pyrogens. When tested as specified in 4.4.5, syringes and needles shall comply with the pyrogen test requirements of the USP.

3.8 Sterility. The syringes and needles shall be sterile when tested with a biological indicator as specified in 4.4.6.

3.8.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.8.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.9 Certificates. For destination inspection only, certificates of freedom from pyrogens, and safety, as specified below, shall be submitted by the contractor to Defense Personnel Support Center, Directorate of Medical Materiel, ATTN: DPSC-ATQ, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, and one copy of each shall be forwarded with each shipment to the consignee.

3.9.1 Certificates of safety (toxicity). A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.6.1 and 4.4.4.1 and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.9.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.7 and 4.4.5, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.10 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile syringes and needles, as applicable, from the containers and preparing them for use. In addition, information as to methods for destroying syringes and needles, as applicable, to prevent their re-use shall also be shown.

* 3.11 Freedom from loose or detachable flash. When tested as specified in 4.4.3.2, flash at the fluid path of the syringe shall not be detachable.

3.12 Workmanship. Workmanship shall be first class throughout. Syringes (including components) shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing determination by technical means of physical and chemical properties) of the item.

* 4.1.2 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling for lot acceptance.

4.2.1 Needles. Inspections shall be performed as specified in MIL-N-36124.

4.2.2 Syringes.

4.2.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105 and, unless otherwise specified, shall be based on the unit of product being one syringe (end item).

4.2.2.2 Sampling for examination. A sample of syringes (and needles) shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptance Quality Level (AQL) shall be as indicated in table II and applies to the syringes only. Refer to MIL-N-36124 for the C of D's on needles.

Table II. Sampling for examination

Type defect	Inspection level 1/	AQL (percent defective)
Major	I	1.0
Minor	I	6.5

1/ Use double sampling plan for normal inspection.

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* 4.2.2.3 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table III.

Table III. Sampling criteria for tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Barrel	Graduation markings	3.1.1.1	4.4.7	S-1	2.5
Plunger tip	Inertness	3.1.2.1	4.4.8	S-2	0.25
Barrel	Dimensions	3.2.1	------	S-1	1.0
Syringe	Accuracy	3.3	4.4.1	S-1	2.5
Syringe	Pressure test	3.4.1 & 3.4.2	4.4.2.1	S-2	2.5
Syringe	Suction test	3.4.3	4.4.2.2 & 4.4.2.3	S-3 1/	1.5
Syringe 2/	Foreign matter	3.5	4.4.3	S-2	4.0
Needle	Corrosion resistance	3.1.3	MIL-N-36124	S-2	1.0
Syringe and needle	Safety	3.6.1	4.4.4.1	S-1	3/
Syringe	Lead	3.6.2	4.4.4.2 and 4.4.4.3	S-2	1.0
Syringe and needle	Freedom from pyrogens	3.7	4.4.5	S-1	3/
Syringe and needle	Sterility	3.8	4.4.6	S-3	3/
Syringe	Flash	3.11	4.4.3.2	S-2 4/	1.0

1/ A minimum of 20 syringes from each lot shall be tested regardless of lot size.
2/ Type II syringes shall be tested with needle.
3/ Acceptance number shall be zero for all sample sizes.
4/ Samples shall be selected at random from those syringes with flash found in the sample group used in the examination for C of D 119.

4.2.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more characteristics.

* 4.3 Examination. The syringes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables II and IV. Examination, however shall not be restricted to defects listed in table IV. Any syringes in the sample containing one or more defects shall be rejected and, if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table I. Classification of defects in accordance with MIL-STD-105

Examine	Defects	
Category		
Visual		
101	Protector missing or dislodged.	
102	Needle missing or dislodged (type II only).	
103	Type I type protector not free of holes, or cotton missing (less than adequate amount) from open type protector.	
104	Tip protector frozen to needle or syringe.	
105	Needle frozen to syringe tip (type II only-forceps may be used to grasp for frozen needle).	
106	Barrel, plunger tip, or both, missing.	
107	Barrel, or plunger tip frozen in barrel.	
108	Inside of barrel not coated with required lubricant.	
109	Inside of barrel contains excessive lubricant.	
110	Barrel tip orifice missing or blocked.	
111	Graduation marking, incomplete, missing, illegible, or incorrect.	
112	Barrel not clear, plunger tip not clearly visible making accurate alignment with graduation markings difficult or impossible.	
113	Edges of barrel tip not smoothly finished, free of pits, tool marks, bubbles, or loose or detachable flash and burs.	
114	Inside surface of barrel not fine uniform surface, free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect fluid tightness and smoothness of its fit with the plunger.	
115	Plunger tip not smooth, free from flash, bubbles, breaks, cracks, or other defects affecting its fit with plunger.	
116	Plunger does not operate smoothly in barrel.	
117	Plunger tip detaches from plunger rod when pushed back and forth in syringe barrel.	
118	Barrel or plunger (including tip) not free of cracks, breaks, crazing or brittleness.	
119	Bore of syringe tip not free from flash which obstructs bore of tip below the minimum permissible area.	

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Table IV. Classification of defects in accordance with MIL-STD-105 (continued)

Minor	
201	Deep scratches (not interfering with reading of dosage, see 112) on outside surface of barrel (except on tapered portion of tip, see 113).
202	Contractor's name or registered trademark, missing, incomplete, or not legible.
203	Graduation scale not approximately in line with flattened side of flange.
204	Instructions missing, incomplete, or not legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy test. Syringes shall be tested for accuracy of delivery at each main graduation mark (see Table I) using water and a calibrated burette, or as an alternate, gravimetrically taking into account the temperature and density of water. The first ring from the distal end of the plunger tip shall be aligned with each graduation mark tested. One-cc syringes shall not be tested for accuracy at the first 15% of the scale.

4.4.2 Fluid tightness test. The fluid tightness shall consist of the pressure test and the suction test described in the following paragraphs:

* 4.4.2.1 Pressure test. For those syringes which come equipped with needles, the following test method shall apply: The water pressure test system shall have a water pressure gage and a medication injection plug. The needle shall be secured to the Luer tip of the syringe by applying it with a 3-1/2 pound axial force and while maintaining this compressive stress twisting the needle hub 45° relative to the Luer tip. This method of attachment does not apply to syringes with permanently affixed needles or syringes with Luer locks. The needle of the hypodermic shall be inserted into the medication injection plug and the water pressure brought up to 45 psig. The plunger shall then be withdrawn to the 50% point of the graduated capacity where it shall be allowed to remain for 10 seconds. The plunger shall then be withdrawn to the 75% point and held there for another 10 seconds. The plunger shall then be withdrawn to the 100% point for a final 10 seconds. During each 10-second period there shall not be enough water to pass the connection between the needle hub and Luer tip of the syringe or needle-syringe junction and between the plunger tip and barrel of the syringe to produce a falling drop. After each successive test, any water that may have passed shall be wiped with a dry cloth and the next test applied. The needle of the hypodermic shall be withdrawn from the medication injection plug and the water ejected at the rate of 1/2 cc/second until the plunger reaches the zero mark. The needle shall not leak or separate at its junction with the syringe.

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4.4.2.2 Suction test. An amount of water equal to more than one-half the capacity shall be drawn into the syringe through a hypodermic needle. The syringe shall then be held vertically, needle up, to allow air to rise above the water and be expelled. Tapping the barrel of the syringe will facilitate removal of trapped air. The plunger shall be pressed inward to the 50% mark. A soft rubber stopper shall then be used to seal the needle. The plunger shall then be carefully withdrawn, at a rate of about one inch/10 seconds, to the full mark. After the plunger has reached the full mark, there shall be no more than three bubbles coming by the distal ring for at least 10 seconds. In testing the Size 7 syringe the plunger shall be withdrawn to the 35cc mark only. Should there be more than three air bubbles passing the distal ring during the 10-second period, there shall be an additional period of observation of 15 seconds during which there shall be no bubbles passing the distal ring. Upon release of the plunger, it shall return to the 50% mark.

Air bubbles which appear on the distal surface of the plunger tip during the withdrawal of the plunger shall be disregarded. Some air bubbles frequently appear under conditions of reduced pressure because of phenomena such as decreeased solubility of air in water, interfacial tension, and aggregation of minute bubbles into larger ones.

4.4.2.3 Alternate suction test. As an alternate, the suction test may be performed with a leak detector which is basically a thermistor conductivity meter using helium. Gas (helium) leakage past the plunger shall be indicated by deflections of the milliammeter needle as listed below for the various sizes of syringes. Suction within the syringe shall be created by withdrawal of the plunger at a rate of approximately one-quarter inch per second. The thermistor conductivity meter which shall be used is the Gow-Mac Model 21-110, or similar apparatus. To minimize hand operation a fixture shall be devised to hold the syringe, couple it to a female Luer fitting, and work the plunger in a manner coincidental with the center line of ths syringe barrel.

Needle Deflection Characteristics (Gow-Mac 21-110)

Size	Sensitivity setting
1cc	Medium
3cc	Medium
5cc	Low
10cc	Low
20cc 1/	Low
30cc 1/	Low
50cc 1/	Low

Full scale deflection of the needle (50 divisions) for two or more seconds shall be considered as leakage and cause for rejection.

1/Momentary full scale defelction shall not be considered an indication of leakage.

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4.4.3 Freedom from foreign matter test. Syringe shall be tested as removed from individual container. Type II syringe shall be tested with the needle attached to syringe. The syringe shall be first examined with the naked eye for obvious foreign matter and then shall be submitted to the tests described in the following paragraph:

4.4.3.1 Test method. Take an amount of distilled water equal to the capacity of the syringe and pour it into the barrel of the syringe being tested. Insert the piston and eject the distilled water into a medium texture, quantitative filter paper properly inserted into a 60°, 75 mm, smooth glass funnel. As a blank, take an equal amount of distilled water and pour it into a similar, separate funnel filter paper arrangement as used for the syringe. Examine the residue on the filter papers, after the water has passed through them, and determine if foreign material ascribbable to the syringe and needle is present.

* 4.4.3.2 Freedom from loose or detachable flash. Flash at the bore of the tip shall be flexed with a blunt probe, of no more than .015 inch diameter, to determine that flash if present when examined at 7X will not become detached in use. Flexing shall be for three cycles. Each cycle is one back and forth full flexure.

4.4.4 Safety (toxicity) and lead content tests.

4.4.4.1 Safety (toxicity). The finished syringes shall be tested for safety in accordance with the current edition of the USP, for Transfusion and Infusion Assemblies, except that a minimum of 5 samples shall be used to prepare the extract. Where more samples are indicated by Table III they shall be used in multiples (or near multiples) of 5 to prepare a single extract.

4.4.4.2 Test for lead. Ten (10) Gms of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffle furnace of 700°C. to a white ash. The ash shall be boiled for 15 milliliters (ml) of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the Pharmacopeia of the United States, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the USP test. If the sample solution is darker than the limit, small aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.4.3 Alternate test for lead. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnance at 650°C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deinoised water and dilute to 50 ml. Prepare standard lead solutions of 1, 2 and 5 ppm. lead.

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Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and a 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead}$$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Pb)}}{\text{absorbance}}$ 1/

W = Sample weight in grams

1/ The three standards are averaged to give ppm/unit A.

4.4.5 Freedom from pyrogens test. The pyrogen test shall be performed as described in the current edition of the USP, for Transfusion and Infusion Assemblies, except that a minimum of 10 samples shall be used to prepare the effluent. The effluent from 10 filled syringes (and needles) is used for each test. Where more samples are indicated by Table III, they shall be used in multiples (or near multiples) of 10 to prepare a single effluent.

4.4.6 Sterility test. The syringes and needles shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table III, Sampling Criteria for Tests.

4.4.6.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

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4.4.6.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6 Sterility test and 4.4.6.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.6.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.6.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.7 Permanency of graduation markings. The graduation markings shall withstand five vigorous rubbing cycles of a 70% isopropyl alcohol moistened cotton swab. A cycle shall be one up and one down stroke along the length of the barrel. The markings shall remain legible.

4.4.8 Plunger tip inertness. The rubber plunger tip shall be removed from the plunger rod, wiped with a clean, dry cloth to remove silicone lubricant, and immersed in a test tube under a volume of pH 10.5 buffer solution1/ equal to the capacity of the syringe. 2/ The exposure shall be for one hour at 68° to 77° F. (20° to 25° C). There shall be no color or any other visible change of the buffer solution. A control shall be used during the exposure consisting of the same amount of buffer solution in a test tube which does not contain a plunger tip.

1/ The pH 10.5 buffer solution is prepared by dissolving 35.0 gm of dibasic potassium phosphate in 1000 ml of distilled water, and adding 2 ml of 10N potassium hydroxide. Adjust the pH with 18N phosphoric acid or 10N potassium hydroxide to 10.5 ± 0.1. The chemicals specified shall be ACS Regent Grade quality.

2/ On large size plunger tips it is permissible to cut them into quarters and use only 1/4 volume of buffer solution.

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4.4.9 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table V. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing fifty or one hundred syringes of the type and size, as specified, consitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI. shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI. Packaging quantities

Type	Size	Unit package	Intermediate package	Exterior container
I	1 and 2	1 unit	10 units	20 units
I	3 and 4	1 unit	5 units	10 units
I	5 and 6	1 unit	4 units	8 units
I	7	1 unit	Not required	Not required
II	1, and 2	1 unit	10 units	20 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Individual container. Each syringe (type I or type II shall be packaged in an individual sealed, peel type container capable of protecting the needle cannula as well as maintaining sterility of contents unless the package is damaged or opened. The peel type containers shall be designed with tabs to facilitate dispensing of the syringe by sterile techniques. As an alternate, the syringe may be packaged in a rigid plastic tamperproof container so designed as to facilitate dispensing the syringe by sterile techniques and to maintain sterility of syringe unless the container is damaged or opened.

5.1.3.2 Unit package.

5.1.3.2.1 Type I or II, Size 1 thru 6. Unit package shall be a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, class domestic, or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2.2 Type I, Size 7. One hundred individual containers shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the box specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution. Individual packaged syringes may be packed in units of 20s prior to insertion in the unit package.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be levels A, B or C as specified (See 6.1).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Type I, size B syringes packaged as specified in 5.1.3 shall require no further packing.

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5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in the appendix of PPP-B-636. Type I, size 7 syringes packages as specified in 5.1.3 shall require no further packing.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

* 5.3.1 Individual container. Labeling shall meet the requirements of the Federal Food, Drug and Cosmetic Act. In addition to any commercial markings, each individual container shall also be marked with the item identification, name and address of the manufacturer or registered trademark, sterility lot (control) number, syringe and needle size, instructions for use and destruction, and the legend: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement. As an alternate, the instructions for use and destruction may be shown on the unit package. When the registered trademark of the syringe is visible through the carrier of the individual container, it shall not be required on the individual container.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, name and address of the contractor and the legend that the syringes are sterile and pyrogen-free. The method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat, C-Cold Sterilization) shall be shown following the level of packaging and date of packaging. Example "A-9/69-G". Final sterilization shall be on or after indicated packaging date. Instructions for use and destruction as specified in 3.10 need not be shown on unit package when shown on the individual container. In addition, a caution sheet shall be inserted into each unit package stating in bold red printing the following legend:

"CAUTION: Recommend that medication which is left in disposable syringes for more than 15 minutes, be examined prior to administration for any change in appearance.
"This caution notice is required by military specifications."

NOTE: Type I, size 7 unit packages shall also be marked with exterior container markings as specified in MIL-STD-129.

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* 5.3.3 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, contract or purchase order, name of contractor and the method of sterilization as specified in 5.3.2 shall be shown.

5.3.4 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Ordering data. Procurements documents should specify the following:

(a) Number and date of this specification.
(b) Type and size required (see 1.2).
(c) Gage and length of needle (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
(e) Applicable National Stock Number.

6.2 This specification is intended to cover hypodermic syringes which are normally procured to meet Military requirements. It does not include all types and sizes as indicated by the title nor all types which are commercially available.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

6.4 This specification covers the following items as listed in the Federal Supply Catalog:

National Stock No.	Item Identification	Specification Designation	
		Type	Size
6515-00-149-0120	SYRINGE AND NEEDLE HYPODERMIC, Disposable, 100 Units Insulin, Long Model, 26 or 27 Gage, 1/2 inch Needle, 1 cc, 100s	II	1
6515-00-149-1206	SYRINGE AND NEEDLE HYPODERMIC, Disposable, 23 gage, 1 inch Needle, 3 cc, 100s	II	2

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National Stock No.	Item Identification	Specification Designation Type	Size
6515-00-168-6913	SYRINGE, LUER, Plastic, Disposable, 50 cc, 100s	I	7
6515-00-462-7348	SYRINGE, HYPODERMIC, Disposable, 3 cc, 100s	I	2
6515-00-719-7217	SYRINGE AND NEEDLE, HYPODERMIC, Disposable, 25 Gage, 5/8 inch Needle, 3 cc, 100s	II	2
6515-00-724-4603	SYRINGE, HYPODERMIC, Disposable, 30 cc, 50s	I	6
6515-00-724-4606	SYRINGE, HYPODERMIC, Disposable, 20 cc, 50s	I	5
6515-00-754-0406	SYRINGE, HYPODERMIC, Disposable, 5 cc, 100s	I	3
6515-00-754-0412	SYRINGE, HYPODERMIC, Disposable, 10 cc, 100s	I	4
6515-00-982-4205	SYRINGE AND NEEDLE, HYPODERMIC, Disposable, Tuberculin, 25 Gage, 5/8 inch Needle, 1cc, 100s	II	1
6515-00-982-4206	SYRINGE, HYPODERMIC, Disposable, Tuberculin, 1 cc, 100s	I	1

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1931
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INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
* MIL-S-36157E, SYRINGES (AND NEEDLES), HYPODERMIC, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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